Exercise 19

Solution 1: Total number of tosses = 500

Number of heads = 285

Number of tails = 215

(i) Let E be the event of getting a head.

$$P(\text{getting a head}) = P(E) = \frac{\text{Number of heads coming up}}{\text{Total number of trials}} = \frac{285}{500} = 0.57$$

(ii) Let F be the event of getting a tail.

$$P(\text{getting a tail}) = P(F) = \frac{\text{Number of tails coming up}}{\text{Total number of trials}} = \frac{215}{500} = 0.43$$

Solution 2: Total number of tosses = 400

Number of times 2 heads appear = 112

Number of times 1 head appears = 160

Number of times 0 head appears = 128

In a random toss of two coins, let E_1, E_2, E_3 be the events of getting 2 heads, 1 head and 0 head, respectively. Then,

(i) $P(\text{getting 2 heads}) = P(E_1) = \frac{\text{Number of times 2 heads appear}}{\text{Total number of trials}} = \frac{112}{400} = 0.28$

(ii) $P(\text{ getting 1 head}) = P(E_2) = \frac{\text{Number of times 1 head appears}}{\text{Total number of trials}} = \frac{160}{400} = 0.4$

(iii) $P(\text{getting 0 head}) = P(E_3) = \frac{\text{Number of times 0 head appears}}{\text{Total number of trials}} = \frac{128}{400} = 0.32$

Solution 3:

Total number of tosses = 200

Number of times 3 heads appear = 39

Number of times 2 heads appear = 58

Number of times 1 head appears = 67

Number of times 0 head appears = 36

In a random toss of three coins, let E_1, E_2, E_3 and E_4 be the events of getting 3 heads, 2 heads, 1 head and 0 head, respectively. Then;

(i) $P(\text{getting 3 heads}) = P(E_1) = \dfrac{\text{Number of times 3 heads appear}}{\text{Total number of trials}} = \dfrac{39}{200} = 0.195$

(ii) $P(\text{getting 1 head}) = P(E_2) = \dfrac{\text{Number of times 1 head appears}}{\text{Total number of trials}} = \dfrac{67}{200} = 0.335$

(iii) $P(\text{getting 0 head}) = P(E_3) = \dfrac{\text{Number of times 0 head appears}}{\text{Total number of trials}} = \dfrac{36}{200} = 0.18$

(iv) $P(\text{getting 2 heads}) = P(E_4) = \dfrac{\text{Number of times 2 heads appear}}{\text{Total number of trials}} = \dfrac{58}{200} = 0.29$

Solution 4: Total number of throws = 300

In a random throw of a dice, let E_1, E_2, E_3, E_4, be the events of getting 3, 6, 5 and 1, respectively. Then,

(i) $P(\text{getting 3}) = P(E_1) = \dfrac{\text{Number of times 3 appears}}{\text{Total number of trials}} = \dfrac{54}{300} = 0.18$

(ii) $P(\text{getting 6}) = P(E_2) = \dfrac{\text{Number of times 6 appears}}{\text{Total number of trials}} = \dfrac{33}{300} = 0.11$

(iii) $P(\text{getting 5}) = P(E_3) = \dfrac{\text{Number of times 5 appears}}{\text{Total number of trials}} = \dfrac{39}{300} = 0.13$

(iv) $P(\text{getting } 1) = P(E_4) = \dfrac{\text{Number of times 1 appears}}{\text{Total number of trials}} = \dfrac{60}{300} = 0.20$

Solution 5: Total number of ladies $= 200$

Number of ladies who like coffee $= 142$

Number of ladies who dislike coffee $= 58$

Let E_1 and E_2 be the events that the selected lady likes and dislikes coffee, respectively.

Then,

(i) $P(\text{selected lady likes coffee}) = P(E_1) = \dfrac{\text{Number of ladies who like coffee}}{\text{Total number of ladies}} = \dfrac{142}{200} = 0.71$

(ii) $P\ (\text{selected lady dislikes coffee}) = P(E_2) = \dfrac{\text{Number of ladies who dislike coffee}}{\text{Total number of ladies}} = \dfrac{58}{200} = 0.29$

Solution 6:

Total number of unit tests $= 6$

Number of tests in which the student scored more than 60% marks $= 2$

Let E be the event that he got more than 60% marks in the unit tests.

Then,

Required probability $= P(E)$

$$= \frac{\text{Number of unit tests in which he got more than 60\% marks}}{\text{Total number of unit tests}}$$

$$= \frac{2}{6}$$

$$= \frac{1}{3}$$

Solution 7: Total number of vehicles going past the crossing $= 240$

Number of two-wheelers $= 84$

Let E be the event that the selected vehicle is a two-wheeler. Then, required probability $= P(E) = \frac{84}{240} = 0.35$

Solution 8: Total phone numbers on the directory page = 200

(i) Number of numbers with units digit 5 = 24

Let E_1 be the event that the units digit of selected number is 5.

$\therefore$ Required probability $= P(E_1) = \frac{24}{200} = 0.12$

(ii) Number of numbers with units digit 8 = 16

Let E_2 be the event that the units digit of selected number is 8.

$\therefore$ Required probability $= P(E_2) = \frac{16}{200} = 0.08$

Solution 9: Total number of students = 40

(i) Number of students with blood group O = 14

Let E_1 be the event that the selected student's blood group is O.

$\therefore$ Required probability $= P(E_1) = \frac{14}{40} = 0.35$

(ii) Number of students with blood group AB = 6

Let E_2 be the event that the selected student's blood group is AB.

$\therefore$ Required probability $= P(E_2) = \frac{6}{40} = 0.15$

Solution 10: Total number of salt packets = 12

Number of packets which contains more than 2 kg of salt = 5

$\therefore$ P(Chosen packet contains more than 2 kg of salt)

$= \frac{\text{Number of packets which contains more than 2 kg of salt}}{\text{Total number of salt packets}}$

$= \frac{5}{12}$

Thus, the probability that the chosen packet contains more than 2 kg of salt is $\frac{5}{12}$

Solution 11: Number of balls played by the batsman $= 30$

Number of balls in which he hits boundaries $= 6$

$\therefore$ Number of balls in which he did not hit a boundary $= 30 - 6 = 24$

P(Batsman did not hit a boundary)

$= \frac{\text{Nmber of balls in which he did not hit a boundary}}{\text{Number of balls played by the batsman}}$

$= \frac{24}{30}$

$= \frac{4}{5}$

Thus, the probability that he did not hit a boundary is $\frac{4}{5}$.

Solution 12: Number of families surveyed $= 2400$

(i) Number of families earning ₹ 25000 – ₹ 30000 per month and owning exactly 2 vehicles $= 27$

$\therefore$ P(Family chosen is earning ₹ 25000 – ₹ 30000 per month and owning exactly 2 vehicles)

= Number of families earning ₹ 25000 – ₹ 30000 per month and owning exactly 2 vehicles / Number of families surveyed

$= \frac{27}{2400}$

$= \frac{9}{800}$

(ii) Number of families earning ₹ 40000 or more per month and owning exactly 1 vehicle $= 579$

∴ P(Family chosen is earning ₹ 40000 or more per month and owning exactly 1 vehicle)

= Number of families earning ₹ 40000 or more per month and owning exactly 1 vehicle / Number of families surveyed

$= \frac{579}{2400}$

$= \frac{193}{800}$

(iii) Number of families earning less than ₹ 25000 per month and not owning any vehicle $= 10$

∴ P(Family chosen is earning less than ₹ 25000 per month and not owning any vehicle)

= Number of families earning less than ₹ 25000 per month and not owning any vehicle / Number of families surveyed

$= \frac{10}{2400}$

$= \frac{1}{240}$

(iv) Number of families earning ₹ 35000 – ₹ 40000 per month and owning 2 or more vehicles $= 59 + 25 = 84$

∴ P(Family chosen is earning ₹ 35000 – ₹ 40000 per month and owning 2 or more vehicles)

= Number of families earning ₹ 35000 – ₹ 40000 per month and owning 2 or more vehicles / Number of families surveyed

$= \frac{84}{2400}$

$= \frac{7}{200}$

(v) Number of families owning not more than 1 vehicle

$=$ Number of families owning 0 vehicle $+$ Number of families owning 1 vehicle

$= 10 + 0 + 1 + 2 + 1 + 160 + 305 + 535 + 469 + 579$

$= 2062$

$\therefore$ P(Family chosen is owning not more than 1 vehicle)

$=$ Number of families owning not more than 1 vehicle / Number of families surveyed

$= \frac{2062}{2400}$

$= \frac{1031}{1200}$

Solution 13: Total number of students $= 30$

(i) Number of students whose marks are 30 or less $= 7 + 10 + 6 = 23$

$\therefore$ P(Marks of the chosen student are 30 or less)

$= \frac{\text{Number of students whose marks are 30 or less}}{\text{Total number of students}}$

$= \frac{23}{30}$

(ii) Number of students whose marks are 31 or more $= 4 + 3 = 7$

$\therefore$ P(Marks of the chosen student are 31 or more)

$= \frac{\text{Number of students whose marks are 31 or more}}{\text{Total number of students}}$

$= \frac{7}{30}$

(iii) Number of students whose marks lie in the interval 21–30 $= 6$

$\therefore$ P(Marks of the chosen student lie in the interval 21–30)

$= \frac{\text{Number of students whose marks lie in the interval } 21 - 30}{\text{Total number of students}}$

$= \frac{6}{30}$

$= \frac{1}{5}$

Solution 14: Total number of teachers = 75

(i) Number of teachers who are 40 or more than 40 years old $= 37 + 8 = 45$

$\therefore$ P(Selected teacher is 40 or more than 40 years old)

$= \frac{\text{Number of teachers who are 40 or more than 40 years old}}{\text{Total number of teachers}}$

$= \frac{45}{75}$

$= \frac{3}{5}$

(ii) Number of teachers of an age lying between 30 – 39 years (including both) = 27

$\therefore$ P(Selected teacher is of an age lying between 30 – 39 years (including both))

$= \frac{\text{Number of teachers of an age lying between } 30 - 39 \text{ years (including both)}}{\text{Total number of teachers}}$

$= \frac{27}{75}$

$= \frac{9}{25}$

(iii) Number of teachers 18 years or more and 49 years or less $= 3 + 27 + 37 = 67$

$\therefore$ P(Selected teacher is 18 years or more and 49 years or less)

$= \frac{\text{Number of teachers 18 years or more and 49 years or less}}{\text{Total number of teachers}}$

$= \frac{67}{75}$

(iv) Number of teachers 18 years or more old $= 3 + 27 + 37 + 8 = 75$

$\therefore$ P(Selected teacher is 18 years or more old)

$= \dfrac{\text{Number of teachers 18 years or more old}}{\text{Total number of teachers}}$

$= \dfrac{75}{75}$

$= 1$

(v) Number of teachers above 60 years of age $= 0$

$\therefore$ P(Selected teacher is above 60 years of age)

$= \dfrac{\text{Number of teachers above 60 years of age}}{\text{Total number of teachers}}$

$= \dfrac{0}{75}$

$= 0$

Solution 15: Total number of patients = 360

(i) Number of patients whose age is 30 years or more but less than 40 years = 60

Let E_1 be the event that the selected patient's age is in between 30 – 40

$\therefore$ P(patient's age 30 is years or more but less than 40 years) $= P(E_1) = \dfrac{60}{360} = \dfrac{1}{6}$

(ii) Number of patients whose age is 50 years or more but less than 70 years

$= (50 + 30) = 80$

Let E_2 be the event that the selected patient's age is in between 50 - 70.

$\therefore$ P(patient's age is 50 years or more but less than 70 years) $= P(E_2) = \dfrac{80}{360} = \dfrac{2}{9}$

(iii) Let E_3 be the event that the selected patient is 10 years or more but less than 40 years.

Number of patients whose age is 10 years or more but less than 40 years $= 90 + 50 + 60 = 200$

$\therefore$ P(Patient's age is 10 years or more but less than 40 years) $= P(E_3) = \frac{200}{360} = \frac{5}{9}$

(iv) Number of patients whose age is 10 years or more

$= 90 + 50 + 60 + 80 + 50 + 30 = 360$

Let E_4 be the event that the selected patient's age is 10 years or more. Then

$\therefore$ P(patient's age is 10 years or more) $= P(E_4) = \frac{360}{360} = 1$

(v) Number of patients whose age is less than 10 years = 0

Let E_5 be the event that the selected patient's age is less than 0.

$\therefore$ P(patient's age is less than 10 years) $= P(E_5) = \frac{0}{360} = 0$

Solution 16: Total number of students = 90

(i) Number of students who gets 20% or less marks = Number of students who gets 20 or less marks = 7

$\therefore$ P(Student gets 20% or less marks)

$= \frac{\text{Number of students who gets 20\% or less marks}}{\text{Total number of students}}$

$= \frac{7}{90}$

(ii) Number of students who gets 60% or more marks

$=$ Number of students who gets 60 or more marks $= 10 + 9 = 19$

$\therefore$ P(Student gets 60% or more marks)

$= \frac{\text{Number of students who gets 60\% or more marks}}{\text{Total number of students}}$

$= \frac{19}{90}$

Solution 17: Total number of electric bulbs in the box $= 800$

Number of defective electric bulbs in the box $= 36$

$\therefore$ Number of non-defective electric bulbs in the box $= 800 - 36 = 764$

P(Bulb chosen is non-defective)

$$= \frac{\text{Number of non defective electric bulbs in the box}}{\text{Total number of electric bulbs in the box}}$$

$$= \frac{764}{800}$$

$$= \frac{191}{200}$$

Thus, the probability that the bulb chosen is non defective is $\frac{191}{200}$.

Solution 18:

(i) Probability of an impossible event $= 0$

(ii) Probability of a sure event $= 1$

(iii) Let E be an event. Then, $P(\text{not } E) = 1 - P(E)$.

(iv) $P(E) + P(\text{not } E) = 1$

(v) $0 \leq P(E) \leq 1$

Multiple Choice Questions (MCQs)

Solution 1: (d) $\frac{4}{5}$

Total number of people surveyed $= 645$

Number of people who have a high school certificate $= 516$

$\therefore$ P(Person has a high school certificate)

$$= \frac{\text{Number of people who have a high school certificate}}{\text{Total number of people surveyed}} = \frac{516}{645} = \frac{4}{5}$$

Hence, the correct answer is option (d).

Solution 2: (c) $\frac{1}{5}$

Explanation:

Total number of students $= 40$

Number of students with blood group AB $= 8$

Let E be the event that the selected student's blood group is AB.

$\therefore$ Required probability $= P(E) = \frac{8}{40} = \frac{1}{5}$

Solution 3: (d) 0

Maximum lifetime a bulb has is 1100 hours. There is no bulb with lifetime 1150 hours.

Solution 4: (c) $\frac{3}{4}$

Total number of children surveyed $= 364$

Number of children who liked to eat potato chips $= 91$

Number of children who do not liked to eat potato chips $= 364 - 91 = 273$

$\therefore$ P(Child does not like to eat potato chips)

$$= \frac{\text{Number of children who do not liked to eat potato chips}}{\text{Total number of children surveyed}} = \frac{273}{364} = \frac{3}{4}$$

Hence, the correct answer is option (c).

Solution 5: (b) $\frac{4}{5}$

Total number of times two coins are tossed = 1000

Number of times of getting at most one head

= Number of times of getting 0 heads + Number of times of getting 1 head

= 250 + 550 = 800

$\therefore$ P(Getting at most one head)

$$= \frac{\text{Number of times of getting at most one head}}{\text{Total number of times two coins are tossed}} = \frac{800}{1000} = \frac{4}{5}$$

Hence, the correct answer is option (b).

Solution 6: (b) $\frac{29}{40}$

Explanation:

Total number of bulbs in the lot = 80

Number of bulbs with life time of more than 500 hours = (23 + 25 + 10) = 58

Let E be the event that the chosen bulb's life time is more than 500 hours.

$\therefore$ Required probability = P(E) = $\frac{58}{80} = \frac{29}{40}$

Solution 7: (c) $\frac{13}{40}$

Total number of students surveyed $= 200$

Number of students who does not like the subject Sanskrit $= 65$

$\therefore$ P(Student chosen at random does not like the subject Sanskrit)

$$= \frac{\text{Number of students who does not like the subject Sanskrit}}{\text{Total number of students surveyed}} = \frac{65}{200} = \frac{13}{40}$$

Hence, the correct answer is option (c).

Solution 8: (c) $\frac{5}{12}$

Explanation:

Total number of trials $= 60$

Number of times tail appears $= 35$

Number of times head appears $= 60 - 35 = 25$

Let E be the event of getting a head.

$$\therefore \text{P(getting a head)} = \text{P(E)} = \frac{\text{Number of times head appears}}{\text{Total number of trials}} = \frac{25}{60} = \frac{5}{12}$$

Solution 9: (b) 0.3

Explanation:

Let E be the event of winning the game. Then,

$\text{P(E)} = 0.7$

$$\begin{aligned} \text{P(not E)} = \text{P(losing the game)} &= 1 - \text{P(E)} \\ &= 1 - 0.7 \\ &= 0.3 \end{aligned}$$

Solution 10: (c) $\frac{4}{5}$

Explanation:

Total number of balls faced $= 30$

Number of times the ball hits the boundary $= 6$

Number of times the ball does not hit the boundary $= (30 - 6) = 24$

Let E be the event that the ball does not hit the boundary. Then,

$$\mathrm{P(E)} = \frac{\text{Number of times ball does not hit the boundary}}{\text{Total number of balls}} = \frac{24}{30} = \frac{4}{5}$$

Solution 11: (b) $\frac{5}{16}$

Explanation:

Total number of cards in the bag $= 16$

Numbers on the cards that are divisible by 3 are 3, 6, 9, 12 and 15.

Number of cards with numbers divisible by $3 = 5$

Let E be the event that the chosen card bears a number divisible by 3.

$\therefore$ Required probability $= \mathrm{P(E)} = \frac{5}{16}$

Solution 12: (b) $\frac{2}{5}$

Explanation:

Total number of balls in the bag $= 5 + 8 + 7 = 20$

Number of black balls $= 8$

Let E be the event that the chosen ball is black.

$\therefore$ Required probability $= \mathrm{P(E)} = \frac{8}{20} = \frac{2}{5}$

Solution 13: (c) $\frac{31}{65}$

Explanation:

Total number of throws = 65

Let E be the event of getting a prime number.

Then, E contains 2, 3 and 5, i.e. three numbers.

$\therefore$ P(getting a prime number)

$= P(E)$

$= \dfrac{\text{Number of times prime numbers occur}}{\text{Total number of throws}}$

$= \dfrac{(10 + 12 + 9)}{65}$

$= \dfrac{31}{65}$

Solution 14: (a) $\dfrac{12}{25}$

Explanation:

Total number of trials = 50

Let E be the event of getting an even number.

Then, E contains 2, 4 and 6, i.e. 3 even numbers.

$\therefore P$(getting an even number)

$= P(E)$

$= \dfrac{\text{Number of times even numbers appear}}{\text{Total number of throws}}$

$= \dfrac{(9 + 7 + 8)}{50}$

$= \dfrac{24}{50}$

$= \dfrac{12}{25}$

Solution 15: (d) $\dfrac{1}{12}$

Explanation:

Total number of students = 36

Number of students born in October = 3

Let E be the event that the chosen student was born in October. Then,

$$P(E) = \frac{\text{Number of students born in October}}{\text{Total number of students}} = \frac{3}{36} = \frac{1}{12}$$

Solution 16: (c) $\frac{11}{15}$

Number of times two coins are tossed simultaneously = 600

Number of times of getting at least one head

= Number of times of getting 1 head + Number of times of getting 2 heads

= 206 + 234

= 440

$\therefore$ P(Getting at least one head)

$$= \frac{\text{Number of times of getting at least one head}}{\text{Number of times two coins are tossed simultaneously}}$$

$$= \frac{440}{600}$$

$$= \frac{11}{15}$$

Hence, the correct answer is option (c).